

QUANTUM KEY DISTRIBUTION FOR ADS-B DATA TRANSMISSION SECURITY IN AIRCRAFT

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Abstract— This research aims to investigate the security concerns of Automatic Dependent Surveillance-Broadcast (ADS-B) communication in the aviation industry and propose the Quantum Key Distribution (BB84) algorithm to generate a symmetric key using the principles of quantum mechanics and use it as a secret key for all encryption/decryption algorithms used in the ADS-B system to increase the security level and avoid any security concerns. ADS-B is a critical technology for air traffic control, providing real-time position and velocity information to air traffic controllers. However, the security of ADS-B communication is a major concern, as it is vulnerable to various security threats, including eavesdropping, data manipulation and injection attacks that could lead to accidents and loss of life. To ensure safe and efficient air travel, A reliable and secure communication system for ADS-B is essential. In this study, we will provide a comprehensive literature review of ADS-B communication and existing encryption methods. We will evaluate the performance of the proposed BB84-Quantum algorithm through simulations and assess its effectiveness in securing ADS-B communication. Our research aims to provide insight into the potential application of the proposed algorithm in the aviation industry. The proposed algorithm's originality lies in its use of quantum mechanics principles to provide secure communication between aircraft and ground stations, ensuring a safe connection for sending and receiving messages in ADS-B packets.

Keywords: *ADS-B; Aviation; BB84-quantum algorithm; Communication security; Encryption; Quantum Key Distribution*

I. INTRODUCTION

The world's skies are busy with millions of aircraft flying daily, and it is crucial to keep track of each of them for air traffic control, safety, and security reasons. Automatic Dependent Surveillance-Broadcast (ADS-B) is a technology that serves this purpose by periodically broadcasting the aircraft's position, velocity, identity, and other critical information up to 6.2 times per second via a radio link. This information helps the ground station and other aircraft in the vicinity to be aware of the location and movement of an aircraft in real-time, making air traffic control easier and more efficient, it is an important technology that relies heavily on the infrastructure and connectivity provided by telecom companies. Its implementation has improved flight safety and air traffic control and has the potential for other applications beyond aviation, The ADS-B technology relies on Global Navigation Satellite Systems (GNSS) and a Mode S transponder on the aircraft to broadcast the aircraft's data. ADS-B is a form of wireless communication technology that operates using radio frequencies in the 1090 MHz and 978 MHz bands. The technology relies on a network of ground

stations, called Automatic Dependent Surveillance Ground Stations (ADS-B GS), to receive and relay the aircraft's position data. It is also capable of transmitting data to other aircraft, making it an excellent communication tool in the skies. The ADS-B system is designed to replace the secondary surveillance radar (SSR) and provide better positioning accuracy via satellite navigation systems, ADS-B technology is an essential component of modern air traffic control, offering numerous benefits and advantages for the aviation industry.[14] While there are challenges and concerns associated with its implementation, ongoing research and development efforts are working to address these issues and ensure the continued success of ADS-B technology in the future, although it is an excellent tool for air traffic control and communication, it is not a secure system, and this makes it susceptible to various cyber-attacks. ADS-B lacks fundamental security measures such as authentication and encryption, making it easy for attackers to forge or tamper with the transmitted aircraft data. As a result, the confidentiality, integrity, and availability of the transmitted aircraft data are at risk.

Cybersecurity threats in aviation have been increasing in recent years, and ADS-B is not exempted from this trend. Various wireless attacks such as jamming, Denial-of-Service (DoS), and spoofing are becoming more common. Attackers could use these methods to disrupt the communication between the aircraft and the ground station, even hijack the aircraft. Other types of attacks on ADS-B systems include eavesdropping, where attackers can intercept and listen to the ADS-B signal, and replay attacks, where attackers can record and replay legitimate ADS-B signals to create fake aircraft signals. The lack of basic security mechanisms in the ADS-B signal makes it easy to forge or tamper with, which affects the confidentiality, integrity, and availability of the transmitted aircraft data. Attackers targeting a specific insecure protocol may formulate many new types of attacks that have not been investigated before in a specific context. Therefore, it is crucial to implement effective security measures to protect ADS-B systems from cybersecurity attacks. Quantum algorithms are rapidly gaining attention in the aviation industry as they offer the potential to revolutionize various applications, from optimizing air traffic control to improving aircraft design and maintenance. One such application is the use of quantum algorithms for secure communication in the aviation industry. With the increasing number of cyber-attacks on aircraft systems, secure communication is a top priority for airlines and aviation companies. Quantum algorithms offer a promising solution to this problem, as they enable secure encryption and decryption of data that is transmitted between

aircraft, ground stations, and air traffic control centers, Quantum key distribution (QKD) is a technique that is used for secure communication using quantum algorithms. QKD allows two parties to share a secret key that can be used for secure communication. The key is generated using quantum states that are transmitted between the two parties. The security of the key is based on the laws of quantum mechanics, which state that any measurement made on a quantum system will disturb it. Thus, if an eavesdropper tries to intercept the communication. The BB84 algorithm is one of the most widely used quantum algorithms for QKD. The algorithm was first proposed by Charles Bennett and Gilles Brassard in 1984. The algorithm is based on the principles of quantum mechanics and allows two parties to establish a secret key that can be used for secure communication.

The security of the BB84 algorithm is based on the laws of quantum mechanics, which state that any measurement made on a quantum system will disturb it. Thus, if an eavesdropper tries to intercept the communication, she will inevitably disturb the quantum states and be detected. The security of the BB84 algorithm depends on the ability to detect the presence of an eavesdropper. Overall, the BB84 algorithm is considered a highly secure algorithm for encrypting and decrypting data that is transmitted by ADS-B that has not yet been widely used in the aviation industry. The algorithm is based on the principles of quantum mechanics. However, the implementation of the BB84 algorithm requires specialized equipment, and the performance of the algorithm may be affected by various factors, such as channel noise and loss. Therefore, careful consideration should be given to the practical aspects of implementing the algorithm in real-world scenarios. At present, classical cryptography techniques are commonly used for encryption in aviation, and while they are typically considered secure, they are susceptible to being hacked by quantum computers. Conversely, quantum cryptography methods, like the BB84 algorithm, are extremely secure and impervious to attacks from quantum computers. Apply the BB84 algorithm in aviation would provide a safe method of communication between aircraft, ground stations, and air traffic control centers. This would make it much more difficult for cybercriminals to intercept and manipulate data transmitted between these parties. In addition, the use of quantum cryptography in aviation would help to ensure the integrity of critical systems such as navigation and communication systems. The potential benefits of implementing the BB84 algorithm in aviation are significant, but there are also challenges that need to be overcome. One of the main challenges is the cost of implementing quantum key distribution systems. These systems require specialized hardware and software, which can be expensive to develop and deploy. Another challenge is the need to ensure the interoperability of quantum key distribution systems with existing communication protocols and infrastructure. This will require significant standardization efforts and coordination between industry stakeholders. Despite these challenges, the potential benefits of implementing the BB84 algorithm in aviation make it an important area of research and development. As the aviation industry continues to evolve and adopt new technologies, it is important to ensure that these technologies are secure and resistant to cyber-attacks. The use of quantum cryptography could play a key role in achieving this goal.

II. PRELIMINARIES

1) ADS-B Packet:

ADS-B packet is the basic unit of data transmitted by an aircraft equipped with an ADS-B transponder, the ADS-B packet contains several fields of information, including the aircraft's GPS-derived position, velocity, altitude, and heading, as well as the aircraft's unique identification code. The packet also contains other data such as the aircraft's flight number, the type of aircraft, and the origin and destination airports, ADS-B packets are transmitted by the aircraft at regular intervals, typically once per second. The packets are broadcast on a specific radio frequency and can be received by ground-based receivers or other aircraft equipped with ADS-B receivers. The system provides more accurate and frequent position information, which helps air traffic controllers to better manage air traffic and improve situational awareness. It also allows pilots to see the position of other nearby aircraft on their cockpit displays, reducing the risk of mid-air collisions.

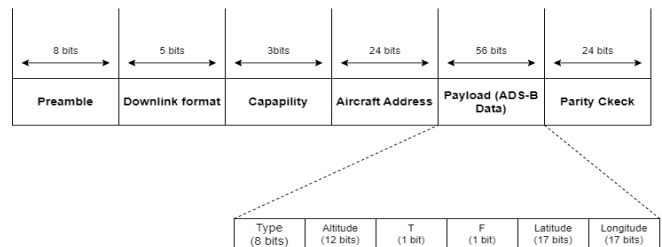


Fig. 1. ADS-B Packet

Fields in ads-b packet are:

- Preamble:** the preamble is a fixed sequence of bits that helps the receiver detect the start of an ADS-B message. The sequence is usually made up of a series of alternating 1s and 0s, which help to synchronize the receiver's clock with the transmitter's clock. This synchronization is important because it ensures that the receiver is able to accurately interpret the message and extract the data contained within it.
- Downlink format:** its field specifies the type of ADS-B message being transmitted. There are several different types of messages, each of which contains a specific set of data fields. For example, a position message might contain information about the aircraft's latitude and longitude, while a velocity message might contain information about the aircraft's speed and direction.
- Capability:** the field specifies the capabilities of the transmitting aircraft. This might include information about the aircraft's ability to communicate on different frequencies, its support for extended squitter technology, or its ability to transmit additional data beyond the standard ADS-B message, then
- Aircraft address:** it contains a unique 24-bit identifier for the transmitting aircraft. This identifier is assigned by the regulatory authority and is used by the receiver to track the aircraft and to ensure message integrity. By assigning a unique identifier to each aircraft, the ADS-B system is able to distinguish between different aircraft and to ensure that the correct data is received and interpreted.

- e) **ADS-B data:** which contains the actual data being transmitted, such as the aircraft's position, velocity, and altitude. The data is formatted according to the downlink format specified in the message. For example, if the downlink format is a position message, the data might include information about the aircraft's latitude, longitude, and altitude.
- f) **Parity check:** is contain a sequence of bits that are used to detect and correct errors in the message. The receiver uses these bits to ensure the accuracy and integrity of the transmitted data. Parity checks are a common technique used in data communications to ensure that the data is transmitted accurately and that errors are detected and corrected.

2) Quantum Key Distribution (BB84) Algorithm:

The proposed algorithm, the BB84 protocol is a quantum key distribution (QKD) protocol, introduced by Charles Bennett and Gilles Brassard in 1984.

The main goal of this protocol is to allow two parties to communicate securely over an insecure channel. In other words, BB84 ensures that no eavesdropper can intercept or read the transmitted messages between the two parties.

The protocol works by using the principles of quantum mechanics to establish a shared key between the two parties, Alice and Bob (sender and receiver). The established key is used to encrypt and decrypt messages, ensuring that only Alice and Bob can read the messages.

The BB84 protocol involves the transmission of photons in different polarizations, which represent 0s and 1s of the key, for the BB84 protocol, we will define the polarization of 0° on the rectilinear basis, or 45° on the diagonal basis as binary 0. Similarly, a binary 1 can be 90° on a rectilinear basis and 135° on a diagonal basis.

- Polarization:

The polarization of a single photon is described by its state vector, which is a superposition of two possible polarization states, typically denoted as $|H\rangle$ (for horizontal polarization) and $|V\rangle$ (for vertical polarization).

The state vector of a photon can be represented as a linear combination of the two polarization states as:

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle$$

where α and β are complex numbers that satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

The elements of the polarization density matrix describe the probabilities of finding the photons in different polarization states. For example, the probability of finding a photon in the horizontal polarization state is given by:

$$P(H) = \langle H|\rho|H\rangle = |\alpha|^2$$

Similarly, the probability of finding a photon in the vertical polarization state is given by:

$$P(V) = \langle V|\rho|V\rangle = |\beta|^2$$

The degree of polarization of the beam is given by:

$$P = (P(H) - P(V)) / (P(H) + P(V))$$

This formula gives the difference in the probabilities of finding the photons in the horizontal and vertical polarization states, normalized by their sum.

The polarization basis we are using to implement the algorithm:

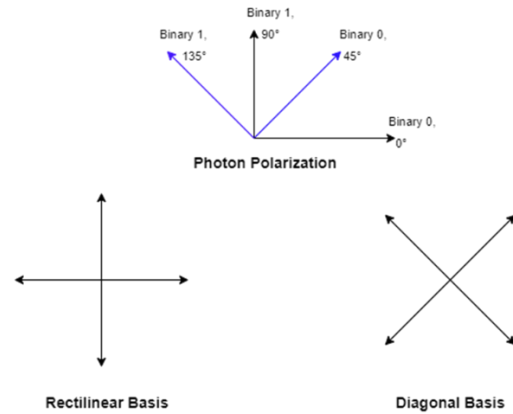


Fig. 2. Polarization Basis

The BB84 protocol consists of the following steps:

1. **SELECT ENCODING:** Alice randomly selects a basis ($\times$ or $+$) to encode each bit.
2. **SELECT MEASUREMENT:** Bob randomly selects a basis ($\times$ or $+$) to measure each bit.
3. **ENCODE:** Alice creates the quantum states, encoded in the selected bases.
4. **SEND:** Alice sends Bob the encoded states, via the quantum channel.
5. **MEASURE:** Bob measures all the quantum states in his pre-selected measurement bases.
6. **ANNOUNCE BASIS:** Alice announces which basis she used to encode each bit, via the classical channel.
7. **FIND SYMMETRIC KEY:** Alice and Bob discard bits in their key that use a different encoding and decoding basis.

The expected output of the algorithm in the absence of any eavesdropper:

TABLE I. Shifted Key Generation

Alice's bit-string	1	0	0	1	1	0	1	0	0	1
Alice's encoding basis	+	X	X	+	X	+	+	+	+	+
Alice's polarization	V	D	D	V	A	H	H	H	H	H
Bob's measurement basis	+	+	X	+	+	X	+	+	+	X
Bob's polarization	V	V	D	V	H	A	H	H	H	D
Shifted Key	1	1	1	1	0	1	0	0	0	1

The expected result is the symmetric key which will be used in the encryption and decryption method as a secret key.

III. LITERATURE REVIEW

- In the age of aviation, the safety and security of passengers and aircrafts are of utmost importance. However, recent studies have identified security vulnerabilities associated with the (ADS-B) data transmissions in aircraft. In a paper published by Jin et al. in 2019, the authors explore these vulnerabilities and propose a timestamp-based security solution to address them, the authors first identify various security threats, including message replay attacks, message spoofing, and location disclosure attacks. These attacks could potentially compromise the safety of passengers and aircrafts. For instance, message spoofing could lead to a false identification of an aircraft, resulting in a collision or other safety issues, to address these vulnerabilities, the authors suggest adding a timestamp to each ADS-B message and validating it on the receiver side. This paper provides valuable insights into the security challenges associated with ADS-B data transmission in aircraft and proposes a solution that can improve the safety and security of the aviation industry. Our team found this paper particularly useful in improving our algorithm implementation as we work towards covering more vulnerability points. By implementing the proposed solution, we can detect and prevent potential security threats, and ultimately, ensure the safety and security of passengers and aircrafts.[1].
- ADS-B signals are critical for aviation safety, providing important information for communication and navigation between aircraft and ground stations. However, decoding these signals can be a challenge due to factors such as channel noise and interference. In a recent study, Jin et al. proposed an optimum receiver for decoding ADS-B signals, the authors provide valuable insights into the signal processing techniques required for decoding ADS-B signals, and the proposed optimum receiver offers a promising solution to the challenges involved. Our team found this paper particularly helpful in enhancing our understanding of signal processing techniques for ADS-B decoding, and we have applied some of the proposed techniques to improve our algorithm's performance. Furthermore, the proposed optimum receiver has the potential to improve the encryption and decryption performance using the BB84-quantum algorithm, which can enhance the security of the data transmission. Overall, the study offers important contributions to the field of aviation safety and security by proposing an innovative solution for ADS-B signal decoding.[9].
- The BB84-quantum algorithm is a game-changing paper in the field of quantum cryptography. The paper introduces a new quantum key distribution protocol that has practical applications in secure communication systems. The BB84 protocol provides a comprehensive explanation of how to securely exchange a secret key between two parties, even in the presence of an eavesdropper. The protocol is based on the principles of quantum mechanics, which offer unique advantages compared to classical key distribution methods, the

authors provide a detailed description of the BB84-quantum algorithm and its practical applications. They explain how the protocol can detect eavesdroppers and prevent information leakage, which makes it an effective solution for secure communication systems. Through the BB84 paper, we have gained valuable insights into the development of efficient and secure quantum cryptography protocols, BB84-quantum algorithm has significant potential in enhancing the security of ADS-B data transmission. It offers an efficient and scalable approach to improve the security of data transmission. By studying this paper, we have come to understand the underlying principles of quantum key distribution and its potential in enhancing the security of ADS-B data transmission. This paper has encouraged us to explore the application of the BB84-quantum algorithm in our project, providing a potent solution that can significantly improve the security of ADS-B data transmission.[3].

- In the world of quantum cryptography, they proposed a new Quantum Key Distribution (QKD) protocol. This protocol was both efficient and secure against eavesdropping attacks and attacks by malicious Bobs. It could be used to establish a secure key exchange between two parties, thereby enhancing the security of ADS-B data transmission in aircraft.
- the paper provided valuable insights into the development of efficient and secure QKD protocols. By reading this paper, a deeper understanding of how QKD works and its potential to improve the security of ADS-B data transmission was gained. The proposed QKD protocol was a powerful solution that motivated further exploration of its application in projects. Moreover, the proposed QKD protocol offered an efficient and scalable approach that could enhance the security of ADS-B data transmission. By using this protocol, a secure and reliable key exchange could be established that could withstand malicious attacks, ensuring the confidentiality and integrity of ADS-B data, the insights gained from this paper could be used to develop more efficient and secure communication systems in the future. The story of this paper reminds us of the importance of constantly exploring new solutions to improve the security of our communication systems.[4]
- The ADS-B systems use insecure communication channels to send and receive the data, making it vulnerable to eavesdropping, signal interference, and lacking privacy. To overcome these weaknesses, we proposed using QKD (BB84) to provide secure communication over these insecure channels. Using quantum channels can help detect any eavesdropping or interference to the channels, as quantum channels are sensitive to any external interaction.

IV. METHODOLOGY

1) BUILDING THE QUANTUM CIRCUIT:

Choosing Bases and Encoding States. Where, the ADS-B Sender generates two binary strings:

One encodes the basis for each qubit:

- 0 $\rightarrow$ Computational basis
- 1 $\rightarrow$ Hadamard basis

The other encodes the state:

- 0 $\rightarrow$ $|0\rangle$ or $|+\rangle$
- 1 $\rightarrow$ $|1\rangle$ or $|-\rangle$

And the ADS-B Receiver also generates a binary string and uses the same convention to choose a basis for measurement. Now that both sides choose their bases and the states has been encoded, we start creating the quantum circuit.

❖ Creating the Circuit:

Based on the following results:

$$\begin{aligned} X|0\rangle &= |1\rangle \\ H|0\rangle &= |+\rangle \\ HX|0\rangle &= |-\rangle \end{aligned}$$

Our algorithm to construct the circuit is as follows:

1. Whenever ADS-B Sender wants to encode 1 in a qubit, he applies an X gate to the qubit. To encode 0, no action is needed.
2. Wherever he wants to encode it in the Hadamard basis, he applies an H gate. No action is necessary to encode a qubit in the computational basis.
3. He then _sends_ the qubits to ADS-B Receiver (symbolically represented in this circuit using wires).
4. The ADS-B Receiver measures the qubits according to his binary string. To measure a qubit in the Hadamard basis, he applies an H gate to the corresponding qubit and then performs a measurement on the computational basis.

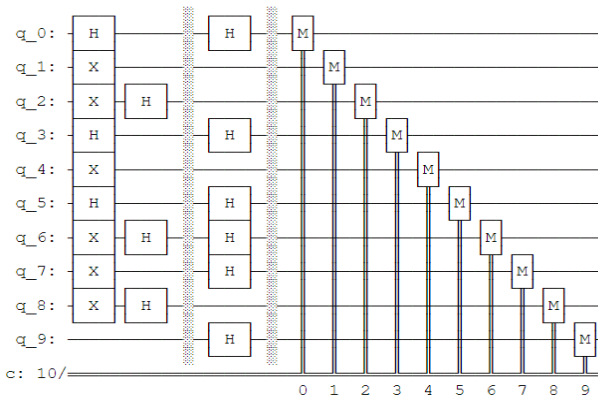


Fig. 3. BB84 Quantum circuit

BB84 quantum circuit of 10 qubits.

2) SYMMETRIC KEY ESTABLISHMENT:

Before sending the ADS-B Packet through the communication channel, we generate the symmetric key that will be used in the Encryption and Decryption processes.

Using the Quantum Key Distribution (BB84) Quantum algorithm to Generate the symmetric key as shown in the following diagram:

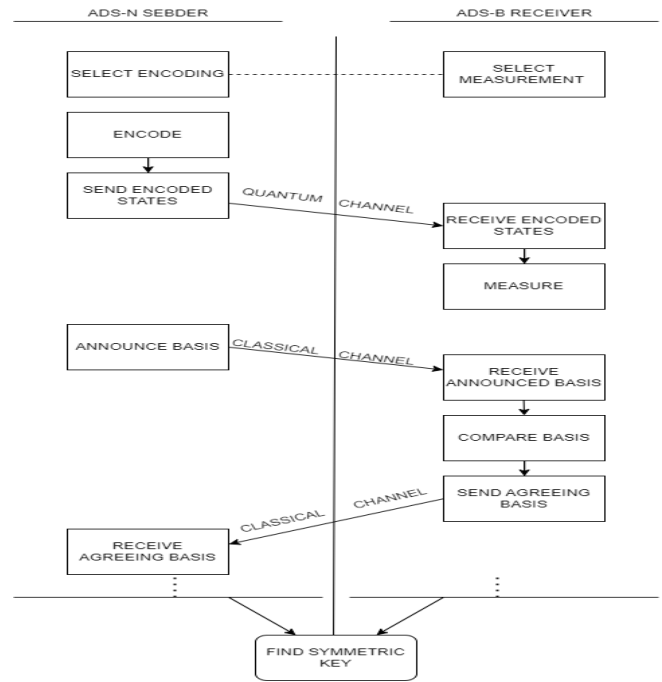


Fig. 4. Symmetric Key Generation

Generating the symmetric key using QKD(BB84), where both sides follow the required steps, Firstly, both ADS-B Sender and Receiver will select their basis randomly.

- **ENCODING:**
After selecting the basis, ADS-B Sender will create the quantum states and encode it in the selected basis.
- **SEND/RECEIVE ENCODED STATES:**
ADS-B Sender will send the encoded states to the ADS-B Receiver through the QUANTUM CHANNEL.
- **MEASURE:**
ADS-B Receiver will measure all the received quantum states in his pre-selected measurements (Basis).
- **ANNOUNCE BASIS:**
ADS-N Sender will announce the basis he used to encode each bit using the CLASSICAL CHANNEL.
- **COMPARE BASIS:**
Upon receiving the announced basis, ADS-B Receiver will compare it with his pre-selected basis.
- **SEND/RECEIVE AGREEING BASIS:**
ADS-B Receiver will send the results of comparing the basis as the "AGREEING BASIS" to the other side through the CLASSICAL CHANNEL.
- **FIND SYMMETRIC KEY:**
Both sides agreed for the final basis, ADS-B Sender and receiver discard bits in their key that used a different encoding and decoding basis, the remaining bits will be the final SYMMETRIC KEY.

3) DATA HANDLING AND DATA TRANSMISSION

The final step of our methodology is to applying the results of the Quantum Key Distribution (which is the symmetric key) on the ADS-B packet.

The following diagram shows the process of handling the ADS-B packet before and after the transmission of the packet:

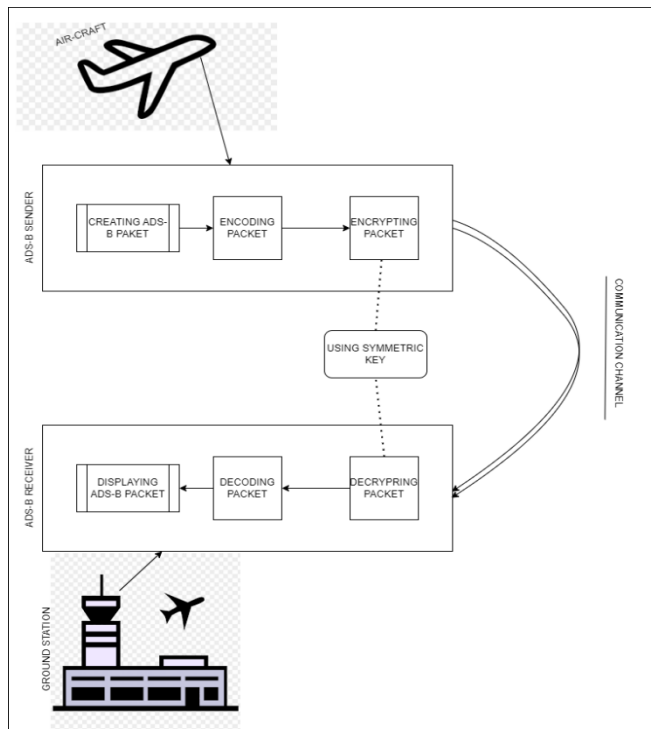


Fig. 5. ADS-B Packet Handling and Transmission

- **CREATING THE ADS-B PACKET:**
The ADS-B packet is being created from the aircraft data, by selecting the required data to the packet, form it in the required size before encoding and encrypting it to be ready to send.
- **ENCODING THE PACKET:**
Encoding the ADS-B packet by converting it to a binary string.
- **ENCRYPTING THE PACKET:**
Using the symmetric key generated by the Quantum Key Distribution (BB84) algorithm as the secret key to perform a bitwise XOR operation between the i^{th} character of two strings: bitstring of the encoded packet and the symmetric key. The result of this operation is the encrypted packet.
- **SINDING THE DATA:**
Sending the encrypted ADS-B packet through the specified communication channel.
- **DECRYPTING THE PACKET:**
By performing the reverse operation of the encryption method, given a string of encrypted bits, it uses the same symmetric key to decrypt the message by performing a bitwise XOR operation between each character of the encrypted bits and the corresponding character of the symmetric key. The result of each XOR operation is converted back to the encoded packet.

- **DECODING THE PACKET:**
Decoding the packet using the same format used for encoding before it was sent, which results in the original ADS-B packet.
- **DISPLAYING ADS-B PACKET:**
The received ADS-B packet has been decrypted and decoded, now it's ready to be displayed at the ground station to check the aircraft status.

V. SYSTEM DESIGN

A. ADS-B System Architecture:

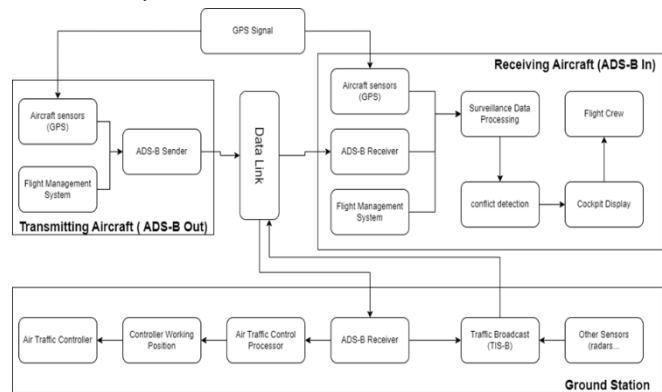


Fig. 6. ADS-B System Architecture

The ADS-B System architecture illustrate the ADS-B data generating, handling, sending, and receiving. The GPS data and other aircraft information are being sent as an ADS-B packet through the communication channel to other aircraft or the ground station, and vice-versa.

The location information is received from the GPS sensors in the aircraft, and other aircraft information is gathered from the flight management system; all this information is sent to the ADS-B report model inside the ADS-B sender which will create the ADS-B packet. The ADS-B packet will be encoded and encrypted before sending to other aircraft or ground station.

The packet will go through a secure channel to reach the ADS-B receiver in the other side (ground station or other aircraft), after receiving the packet it will be decrypted, decoded, and processed in the data processing unit. Then it will be displayed to the Flight-Crew or Air-traffic-Controller.

VI. RESULTS

We test the accuracy of the symmetric key with the absence of any eavesdropping or interception to the communication channel, by following the steps of the algorithm to generate the key.

Running a data integrity test to during the implementation of the algorithm, to ensure the data has not been modified, deleted, or corrupted during the transmission, with a sample of key length equal to 100.

```

python3 main.py --sender-key=0011100110011111100101100001100100001100110011110000111100001000110011110011
alice_bitstring: 00111001100111100101100001100100001100110011110000111100001000110011110011
alice_bases: 0011100110011110010110000110010000110011001111000011001000011001001110010110010000
selected_measurements: 0100110001010101011000000011011001110001011000100011001111111001100000011111
alice_key: 01101100110010011100011011110000111000111000110010101001001110000100010010011000101100
bob_key: 01101100110010011100011011110001110001100101010010110000100010100110001001100
Alice's key is equal to Bob's key: True
One random computer-generated 100-bit symmetric key

```

Fig. 7. Data Integrity Test

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